

Exploring Quantum Generative Adversarial Network with GSoC



Maya Ma

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Over the past few months, I've been experimenting with **Quantum Generative Adversarial Networks (QGANs)**, trying to understand how far quantum circuits can go in generating meaningful data, and what it takes to make them train reliably.

It's been a mix of excitement, confusion, and debugging marathons. Along the way, I built several models, from fully quantum ones to hybrid architectures, and learned a lot about something we often take for granted in deep learning: **how to actually judge a generator's performance.**

Part 1 — The Challenge of Judging a Generative Model

Evaluating a classifier is easy: you look at accuracy. Evaluating a *generator* is much trickier.

A model might output images that look fine, but could still be missing entire parts of the data distribution.

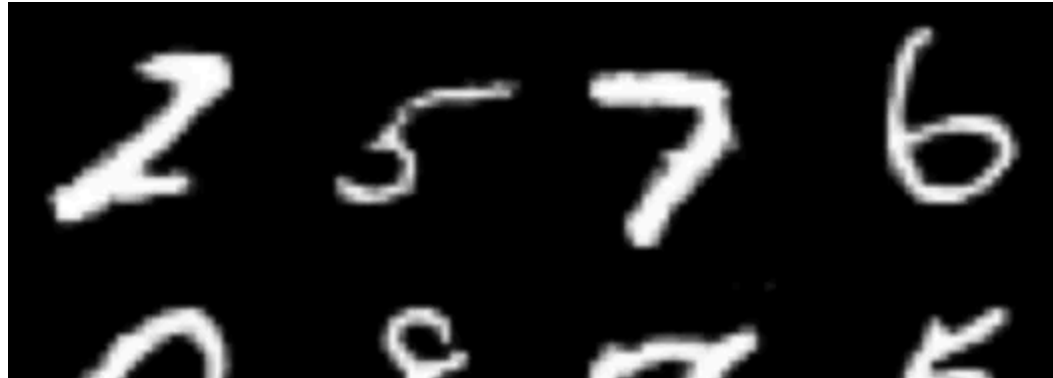
At first, I relied on intuition and visuals. But I quickly realized that “looks good” isn’t a stable metric when training something as delicate as a Quantum GAN. With the help of my mentors, I  Ask AI  nt ways in judging a Generator.

Looking Beyond Visuals

My first step was just visual sanity checks: plotting generated MNIST digits or 4×4 jet images and asking, “Does this make sense?”

That helped spot obvious failures and mode collapse, but not subtle issues like missing diversity or unstable convergence.





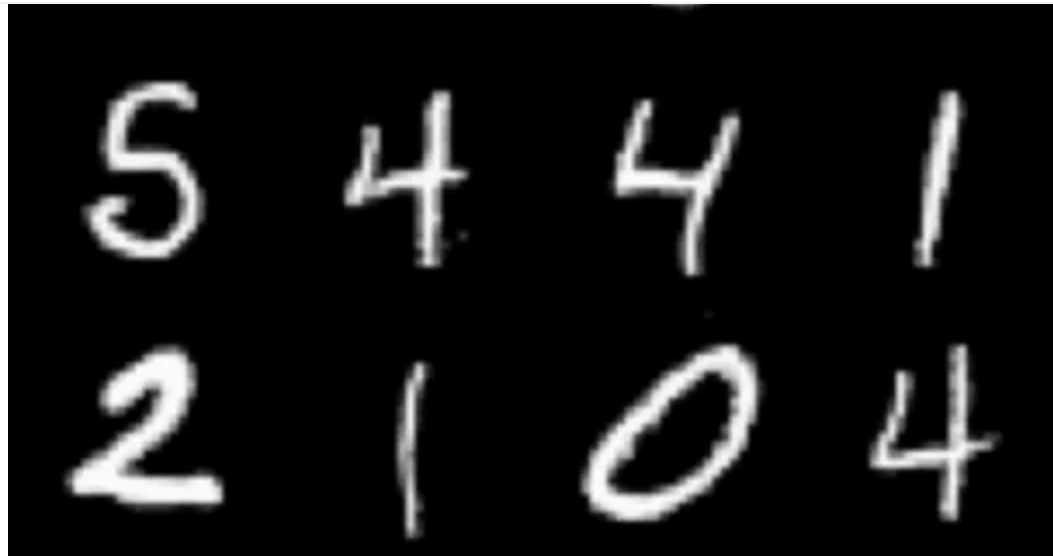
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Quantitative Metrics: FID and IS

Then came the numbers.

I used **Fréchet Inception Distance (FID)** and **Inception Score (IS)**, which are standard for image GANs. They compare distributions of real and fake data



in a deep feature space. Along with other metrics like Mode Collapse Rate, PSNR, and SSIM.

They worked okay for MNIST, but when I switched to my high-energy physics dataset (quark and gluon jet dataset), those metrics stopped being meaningful. Because there was no reliable classifier nor benchmarks.

PCA Histograms: A Hidden Diagnostic

That's when I found something surprisingly effective: **PCA component histograms**.

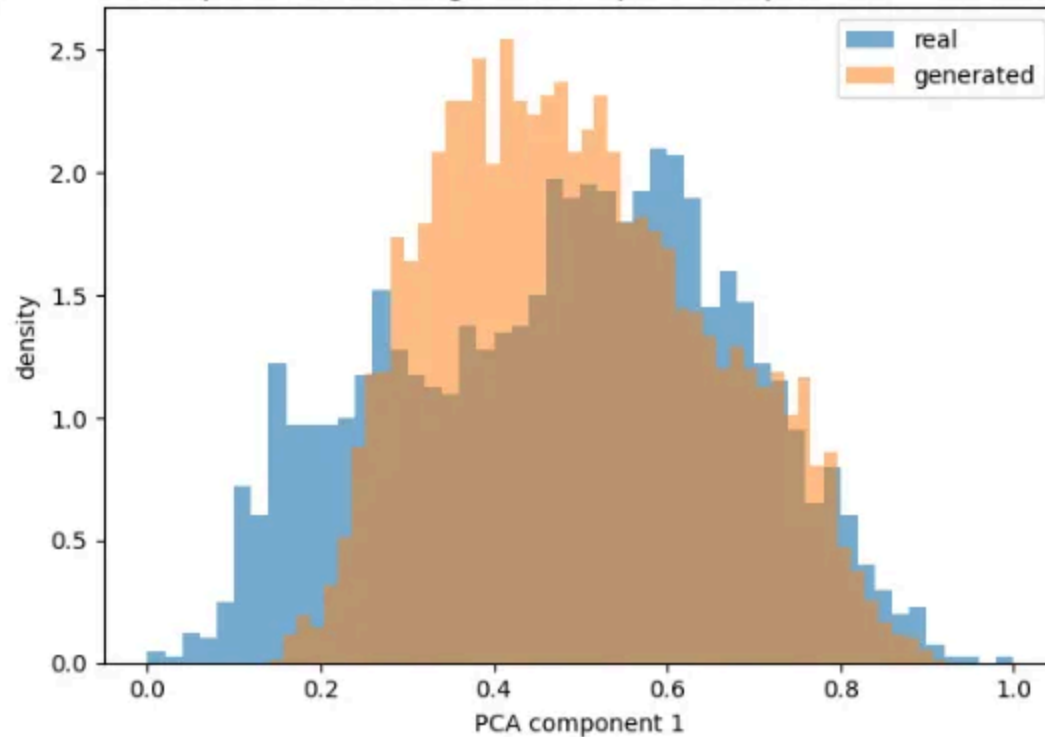
I projected both real and generated samples into their top principal components and plotted their distributions.

When they didn't line up, it was an instant red flag: either the generator had collapsed onto a few modes or the latent sampling wasn't balanced.

After tuning normalization and feature-matching, the PCA plots finally started to align, showing that the generator was covering more of the real distribution.



K_FEAT=12, samples (real=2000, gen=5000) | PCA Component 1 — Real vs Generated



Watching the Gradients

The next big insight came from tracking **gradient magnitudes**.

In many runs, I noticed that the discriminator's gradient norm dropped to near zero long before the generator stabilized.

That meant the discriminator was learning too quickly, making it impossible for the generator to catch up.

This led me to explore more stable loss formulations and regularization strategies. Eventually, that search led me to the **Wasserstein Hybrid Quantum GAN**, which turned out to be the most consistent and interpretable model I trained.



Part 2 — The Wasserstein Hybrid Quantum GAN (WQGAN-GP)

After testing several designs, IQGAN, EQGAN, conditional quantum GANs, the hybrid Wasserstein model stood out. It struck a rare balance between expressive power and trainability.

Architecture Overview

This model blends a **quantum variational layer** into a classical neural network.

Here's how it's structured:

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Generator:

- Starts with a classical MLP that maps noise z into 6 features.
- Passes those features through a **6-qubit Variational Quantum Circuit (VQC)** with data re-uploading and ring entanglement.
- A final MLP reshapes the quantum outputs into a 4×4 image.
- Uses *softplus* activation for non-negative energy maps (since jet images represent energy).

Discriminator:

- A small CNN with **spectral normalization** for stability.
- Includes a **minibatch standard-deviation channel** to encourage diversity.
- Trained with the **Wasserstein loss + gradient penalty (WGAN-GP)** to keep gradients smooth and meaningful.

$$\mathcal{L}_D = -\left(\mathbb{E}[D(x_{\text{real}})] - \mathbb{E}[D(x_{\text{fake}})]\right) + \lambda \cdot \text{GP}$$

Training Details

I trained the model with:

- **5 critic updates** per generator update
- **Lazy gradient penalty** (every 2 steps)
- **Exponential moving average (EMA)** for the generator
- **Feature matching** and **mode-seeking regularizers** to reduce collapse
- **Instance noise** that slowly decays during training to improve distribution overlap early on

Everything was implemented in **PyTorch** and **PennyLane**, with the VQC wrapped as a differentiable `TorchLayer`.

I used `lightning.qubit` for analytic training and `default.qubit` for noisy simulations when needed.

Part 3 — Lessons Learned

1. Evaluation is everything.

Pretty samples can be misleading. Metrics, PCA plots, and gradient curves tell the real story.

2. Hybrid models are the practical (for now).

Fully quantum GANs are beautiful in theory, but current simulators and hardware limit scalability.

Mixing quantum and classical layers gives you the best of both worlds: expressive quantum features with stable classical optimization.

3. Regularization isn't optional.

Spectral norm, gradient penalty, feature matching, and even a touch of noise made the difference between collapse and convergence.

This project taught me how fragile and fascinating generative training can be — and how quantum circuits introduce both new possibilities and new headaches.

The next steps I'm exploring are:

- Scaling the hybrid model to **8–12 qubits**
- Simulating noisy quantum system to see how noise affects the dynamics

Quantum GANs are still an open frontier, but the lessons from this work, about evaluation, balance, and stability, apply far beyond the quantum realm.



Finally, I thank my mentors Amey Bhatuse, Tom Magorsc, Sergei V. Gleyzer, and K.C. Kong for their valuable suggestions and conversations. A Special mention to Sergei V. Gleyzer, Ph.D., for organizing the QMLHEP GSOC.

You can find all my codes [here](#) or [here](#).

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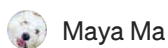
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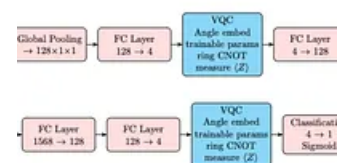
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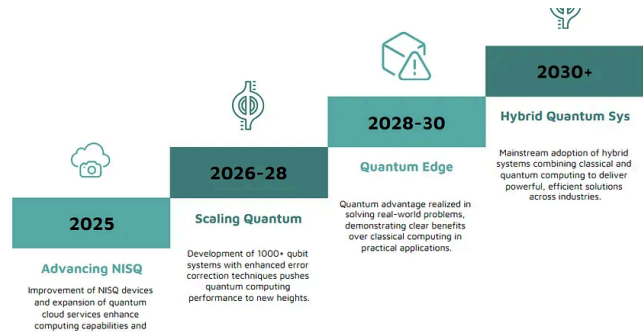
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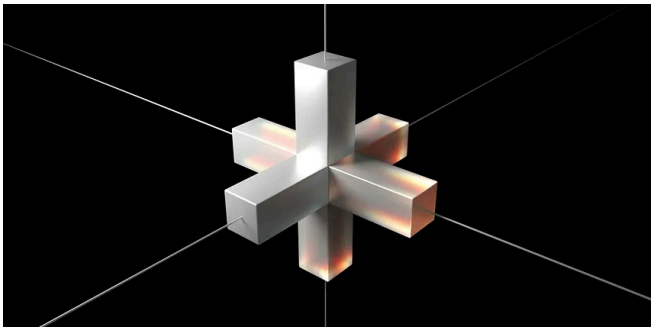


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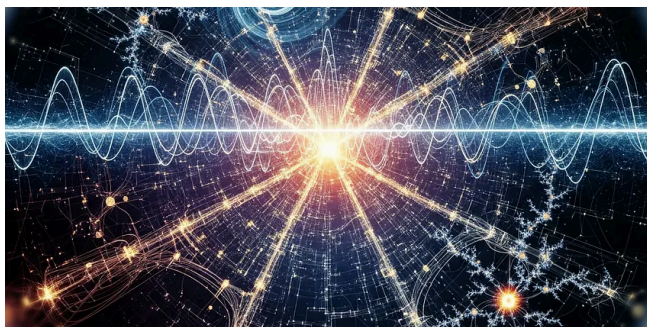
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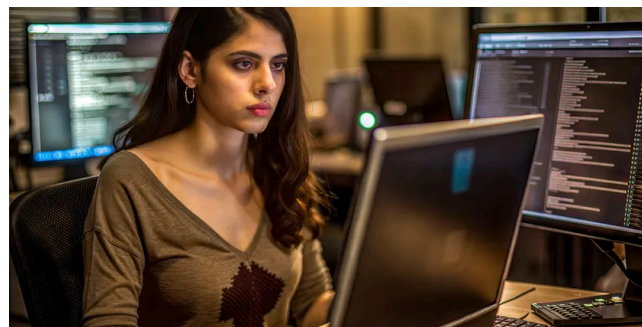
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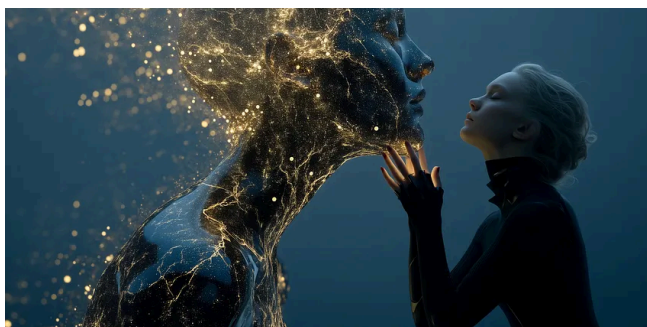
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